Data and Application Architecture

Table of contents

Purpose of this chapter . . . 1
Position in the framework . . . 1
How NSIs use this chapter . . . 2
Requirements and Constraints for Phases C and D . . . 2
Data Architecture . . . 4
- Steady-state datasets . . . 5
- SDMX structural metadata . . . 7
- Governance, quality and design data objects . . . 7

Application Architecture . . . 9
- Application landscape . . . 9
- Application services . . . 17
- Application interfaces . . . 20
- Application services per value stream . . . 21
- Core data platform and SDMX path . . . 21

Purpose of this chapter

The data and application architecture describes which data the statistical organisation keeps, and which applications create, process and deliver that data. It is the layer where the steady states of the business architecture become datasets, and where the business services become application services.

The business architecture is independent of technology. This chapter is the first that is not. It describes a reference solution: a set of data objects, application services and application components that together support the primary value streams VS-01 to VS-06. The components are named with a concrete open-source product, so that an NSI can see what a working solution looks like. The product names are examples. The application services behind them are the stable part of the reference.

The chapter has three parts. The first lists the requirements and the constraint that shape the data, application and technology layers. The second describes the data architecture. The third describes the application architecture.

Position in the framework

In the TOGAF Architecture Development Method, the data and application architecture are developed in Phase C, Information Systems Architectures. In ArchiMate, both belong to the application layer: data objects are the passive elements of that layer, and application components, services and interfaces are the active and behaviour elements.

PLACEHOLDER: Position of the Data and Application Architecture between the Business Architecture and the Technology Architecture

---

Direction	Relationship	Example in the SAF
From Motivation	Requirements realise principles, and are realised by data objects and components	SAF-RQ02 realises SAF-P07 and is realised by the Workflow Orchestrator
From Strategy	Application components realise capabilities	The Statistical Disclosure Control Toolkit realises Statistical disclosure (1.3.7)
From Business Architecture	Data objects realise business objects	*Raw Dataset (#1)* realises the steady state *#1 Raw Data*
From Business Architecture	Application services realise business services and serve business processes	The Data Validation Service realises the Conformance Validation Service and serves the execution process of VS-02
To Technology Architecture	Artifacts realise data objects	The Raw Zone realises *Raw Dataset (#1)*
To Technology Architecture	Technology services serve application components	The Table Storage Service serves the Distributed SQL Query Engine

How NSIs use this chapter

The content of this chapter is reference content at two levels.

Data objects, application services and application interfaces are independent of products. They describe what must exist and what it must do. Most NSIs can adopt them as they are, or map their own landscape onto them.

Application components are named with one open-source product each. This follows SAF-P01 and the constraint SAF-CN01. An NSI uses the component list in one of three ways: it adopts the product, it replaces the product with another one that realises the same application service, or it maps an existing system onto the component. In all three cases the application service stays the point of comparison between NSIs.

An NSI that already has a platform does not need to replace it. It uses this chapter to check whether every application service is covered, by which system, and for which value stream.

Licences are listed with each component. Some products have a copyleft or network-copyleft licence (GPL, AGPL). Under SAF-CN01, these need a legal review before adoption.

Requirements and Constraints for Phases C and D

The data, application and technology layers are shaped by seven requirements and one constraint. Each requirement realises an architecture principle, a goal or an outcome from the Motivation chapter, and is in turn realised by elements of the lower layers. This gives traceability from a principle down to a component or a storage zone.

ID	Requirement	Description	Realises	Realised by
SAF-RQ01	Statistics (#4) and Released Data (#5) are formatted in SDMX	Steady states #4 and #5 follow SDMX. Datasets conform to published Data Structure Definitions, codelists and concept schemes, and are available as SDMX-ML, SDMX-JSON and SDMX-CSV.	SAF-G2, SAF-O3	SAF-DO04, SAF-DO05, SAF-DO06, SAF-DO10, SAF-DO11, SAF-DO12, SAF-AC12, SAF-AC13
SAF-RQ02	Steady-state handover is event-driven and loosely coupled	Streams exchange data only through registered steady-state versions. Registration emits an event that triggers the next stream. No stream calls the internals of another stream.	SAF-P07	SAF-AC04, SAF-AS24, SAF-TSV04
SAF-RQ03	SQL, pipeline and schema definitions are version-controlled	All SQL models, DAGs, schemas, validation suites and policies live in Git. They are reviewed, tested in CI and promoted through environments. Every dataset version records the code version that produced it.	SAF-P03	SAF-DO18, SAF-AC07, SAF-AC23
SAF-RQ04	Every dataset version carries metadata, lineage and BIV classification	Metadata, lineage, quality results and the BIV classification (availability, integrity, confidentiality) are captured automatically at the point of creation and published in the catalogue.	SAF-P09	SAF-DO13, SAF-DO14, SAF-DO16, SAF-AC11

ID	Requirement	Description	Realises	Realised by
SAF-RQ05	Confidentiality is applied progressively and gated before release	Identifiers are pseudonymised at #2, disclosure risk is assessed at #3, initial control is applied at #4, and a final disclosure-control gate precedes #5.	SAF-P11	SAF-DO21, SAF-AC09, SAF-AC19
SAF-RQ06	Access is authenticated, authorized and audited by design	All human and system access goes through single sign-on and policy decisions, and is logged in a tamper-evident audit trail.	SAF-P10	SAF-DO17, SAF-AC18, SAF-AC19, SAF-AC20
SAF-RQ07	Each steady state is a versioned, immutable Iceberg snapshot	A steady state is an Iceberg snapshot tagged as a quality version. Consumers read a tagged version, so that results stay reproducible and shareable.	SAF-P08	SAF-AC06, SAF-AF01, SAF-AF02, SAF-AF03, SAF-AF04, SAF-AF05

ID	Constraint	Description	Realises
SAF-CN01	Open-source components only	The target stack contains only open-source products with OSI-approved licences. Where a licence is copyleft or network-copyleft (GPL, AGPL), legal review is needed before adoption. Alternatives are listed in the component documentation.	SAF-P01

Data Architecture

A **data object** is data structured for automated processing. In ArchiMate, a data object realises a business object: it is the form in which the information of the business is held by applications. A data object is in turn realised by an artifact in the technology layer, which is the physical form in which it is stored.

The SAF data architecture contains 18 data objects in three groups: the steady-state datasets, the SDMX structural metadata, and the data objects for governance, quality and design.

PLACEHOLDER: ArchiMate view *Phase C - Data Architecture: Data Objects and Steady States*

Steady-state datasets

Each steady state of the business architecture is realised by one or two data objects. Each dataset is stored as Apache Iceberg tables in a namespace of its own. The datasets of #4 and #5 are shaped as SDMX (SAF-RQ01).

ID	Data object	Description	Realises business object	Realised by artifact	Accessed by
SAF-DO01	Raw Dataset (#1)	Iceberg tables in namespace `raw`. Data in its original form as received, with provenance, source, method, timestamp and BIV classification. Never edited; each registration is an Iceberg snapshot.	#1 Raw Data	SAF-AF01	SAF-AC01, SAF-AC02, SAF-AC03, SAF-AC05, SAF-AC10
SAF-DO02	Standardized Dataset (#2)	Iceberg tables in namespace `std`. The same values as #1, expressed in common structures, formats and standard classifications. Identifiers are pseudonymised; no personally identifiable information.	#2 Standardized Data	SAF-AF02	SAF-AC05, SAF-AC07, SAF-AC10
SAF-DO03	Processed Dataset (#3)	Iceberg tables in namespace `proc`. Edited, imputed, enriched and integrated microdata with edit lineage, flags and derived variables. Each quality version is an Iceberg tag on a snapshot.	#3 Processed Data	SAF-AF03	SAF-AC05, SAF-AC07, SAF-AC08, SAF-AC10

ID	Data object	Description	Realises business object	Realised by artifact	Accessed by
SAF-DO04	Statistics Dataset - SDMX (#4)	Estimates and aggregates shaped as SDMX (dimensions, measures, attributes), conforming to a published Data Structure Definition, with accuracy measures. Stored in Iceberg namespace `stat`; exportable as SDMX-ML, SDMX-JSON and SDMX-CSV.	#4 Statistics (Output Data)	SAF-AF04, SAF-AF06	SAF-AC05, SAF-AC08, SAF-AC09
SAF-DO05	Statistical Product - SDMX Dataflow (#4)	Approved composite product (indicators, indices, time series), registered as an SDMX Dataflow with categorisation. This is the approved state that VS-05 hands to VS-06.	#4 Statistics (Output Data)		SAF-AC07, SAF-AC12
SAF-DO06	Released Dataset - SDMX (#5)	Disclosure-controlled, documented dataset published as SDMX through the SDMX web service, the open-data portal and research access. Iceberg namespace `rel`.	#5 Released Data	SAF-AF05, SAF-AF06	SAF-AC05, SAF-AC09, SAF-AC13, SAF-AC14, SAF-AC15, SAF-AC16, SAF-AC17

SDMX structural metadata

The SDMX structures are the contract between the producers and the consumers of steady states #4 and #5. Together they realise SAF-RQ01.

ID	Data object	Description	Accessed by
SAF-DO10	SDMX Data Structure Definition (DSD)	SDMX structure that defines the dimensions, measures and attributes of a dataset. It is the contract between producers and consumers of #4 and #5, aligned with GSIM.	SAF-AC12, SAF-AC13
SAF-DO11	SDMX Codelists and Concept Schemes	Standard classifications, code lists and concept definitions (for example NACE and ISCO), used for standardization in VS-02 and in SDMX structures. Versioned and published.	SAF-AC12, SAF-AC13
SAF-DO12	SDMX Dataflow and Categorisation	Catalogue of publishable dataflows organised by category. It is the product and service catalogue for dissemination.	SAF-AC12, SAF-AC13, SAF-AC16

Governance, quality and design data objects

These data objects hold the metadata, logs, policies and definitions that accompany the datasets. Several of them realise business objects of the four process types: design artefacts, provenance and audit logs, and quality reports.

ID	Data object	Description	Realises business object	Accessed by
SAF-DO13	Lineage and Provenance Record	OpenLineage job, run and dataset events, and provenance metadata that capture origin, method, parameters and inputs for every dataset version (SAF-P09).	Provenance, QA & Audit Logs	SAF-AC04, SAF-AC11
SAF-DO14	Data Quality Result	Validation outcomes and quality indicators per dataset version: integrity, conformance, edit impact, accuracy and coherence.		SAF-AC10, SAF-AC11

ID	Data object	Description	Realises business object	Accessed by
SAF-DO15	Data Agreement Record	Register entry per data sharing agreement: provider, legal basis, access rights, SLA and validity.	Data Sharing Agreement	SAF-AC02, SAF-AC03, SAF-AC11
SAF-DO16	Dataset Metadata Entry	Catalogue entry per dataset version: schema, owner and steward, BIV classification, steady state, quality version, and links to lineage and quality results.		SAF-AC06, SAF-AC11
SAF-DO17	Access and Audit Log	Tamper-evident log of authentication, authorization decisions, data access and changes (SAF-P10).	Provenance, QA & Audit Logs	SAF-AC18, SAF-AC19, SAF-AC20
SAF-DO18	Pipeline and SQL Model Definitions	Versioned (Git) Airflow DAGs, dbt SQL models, validation suites, disclosure-control and statistical scripts, and schema migrations. These are the design and implementation artefacts of every stream.	Design Artefacts (Metadata)	SAF-AC04, SAF-AC07, SAF-AC23
SAF-DO19	Quality Indicators and KPI Metrics	Process KPIs (timeliness, rejection rates, conformance, accuracy) and platform metrics that feed quality reports and dashboards.	Quality Report	SAF-AC15
SAF-DO20	Usage and Feedback Data	Web analytics, API usage, user questions and tickets from dissemination (VS-06.5), routed to VS-10.		SAF-AC21, SAF-AC22

ID	Data object	Description	Realises business object	Accessed by
SAF-DO21	Access and Disclosure Policy	Policy-as-code: access rules by BIV classification, role and steady state, and disclosure-control rules and thresholds.		SAF-AC09, SAF-AC19, SAF-AC23

Application Architecture

The application architecture uses three ArchiMate elements. An **application component** is a self-contained unit of software. An **application service** is the behaviour that a component offers to its environment. An **application interface** is the point where that service is made available to users or to other components.

The SAF application architecture contains 23 application components, 24 application services and 5 application interfaces.

Application landscape

The components are grouped into five functional groups.

PLACEHOLDER: ArchiMate view *Phase C - Application Architecture: Landscape*

Acquisition

ID	Component	Product	Licence	Description	Realises service	Realises capability
SAF-AC01	Survey Platform	LimeSurvey	GPL-2.0	Web (CAWI) and self-completion surveys for primary collection. Exports responses and paradata to the raw zone. Field (CAPI) collection can be added with an open-source field tool such as Survey Solutions or ODK.	SAF-AS01	1.2.1 Primary data collection

ID	Component	Product	Licence	Description	Realises service	Realises capability
SAF-AC02	Secure File Transfer Gateway	SFTPGo	AGPL-3.0	SFTP/HTTPS endpoint where providers deliver files. Writes to the landing prefix in S3 and raises a delivery notification. Credentials and quotas per provider.	SAF-AS02	1.2.2 Secondary data collection
SAF-AC03	Ingestion Engine	Apache NiFi	Apache-2.0	Pulls from REST APIs, SFTP, databases and web sources. Records ingestion provenance and lands data in the raw zone. Supports secondary and emerging sources such as web scraping and sensors.	SAF-AS03	1.2.2 Secondary data collection

Processing and Transformation

ID	Component	Product	Licence	Description	Realises service	Realises capability
SAF-AC04	Workflow Orchestrator	Apache Airflow	Apache-2.0	Schedules and runs all pipelines as versioned DAGs (KubernetesExecutor). Asset-aware scheduling starts the next stream when a steady state is registered. Emits OpenLineage events.	SAF-AS04, SAF-AS24	3.2.3 Technical management
SAF-AC05	Distributed SQL Query Engine	Trino	Apache-2.0	Federated SQL over Iceberg tables in S3. Executes standardization, derivation and composition SQL and serves analysts. Access control is delegated to the Policy Decision Engine.	SAF-AS05	1.3 Processing
SAF-AC06	Iceberg REST Catalog	Apache Polaris	Apache-2.0	Table catalog for the namespaces `raw`, `std`, `proc`, `stat` and `rel`. Manages snapshots, branches and tags (quality versions) and table-level access. Alternative: Project Nessie.	SAF-AS06	2.1.1 Data lifecycle management

ID	Component	Product	Licence	Description	Realises service	Realises capability
SAF-AC07	SQL Transformation Framework	dbt Core	Apache-2.0	Version-controlled, tested SQL models for standardization, editing, derivation and composition. Runs on Trino through Airflow. Generates documentation and lineage.	SAF-AS07	1.3.1 Data standardization; 1.3.2 Deriving variables and units
SAF-AC08	Statistical Computing Runtime	R and Python	GPL-2.0 / PSF	Containerised R (survey, VIM, mice) and Python jobs for imputation, weighting and calibration, estimation, and small-area and time-series models. Run as Airflow Kubernetes pods; read and write Iceberg through Trino.	SAF-AS08	1.3.4 Adjusting and imputing; 1.3.5 Estimating population parameters

ID	Component	Product	Licence	Description	Realises service	Realises capability
SAF-AC09	Statistical Disclosure Control Toolkit	sdcMicro, sdcTable	GPL-2.0	Disclosure risk assessment and protection of microdata and tables (suppression, perturbation, k-anonymity). Applied at #4 and as the final gate before #5.	SAF-AS09	1.3.7 Statistical disclosure
SAF-AC10	Data Quality Framework	Great Expectations	Apache-2.0	Declarative expectation suites for intake, conformance, edit and coherence checks. Results are stored as Data Quality Results and published to the catalogue.	SAF-AS10	1.3.3 Check content; 2.1.3 Statistical quality management

Metadata and Governance

ID	Component	Product	Licence	Description	Realises service	Realises capability
SAF-AC11	Metadata and Lineage Platform	OpenMetadata, OpenLineage	Apache-2.0	Dataset catalogue, glossary, BIV classification tags, ownership and column-level lineage, ingested from Airflow, Trino, dbt and OpenLineage. Implements the Metadata Catalog Service.	SAF-AS11	2.1.1 Data lifecycle management; 2.1.2 Metadata management
SAF-AC12	SDMX Structure and Lifecycle Manager	.Stat Suite DLM	MIT (verify per component)	Authoring and registry of SDMX structures (DSD, codelists, concept schemes, dataflows), and loading of statistical data into the dissemination store.	SAF-AS12	2.1.2 Metadata management; 2.1.4 Statistical register management

SDMX and Dissemination

ID	Component	Product	Licence	Description	Realises service	Realises capability
SAF-AC13	SDMX Web Service	.Stat Suite NSI-WS	MIT / EUPL (verify per component)	SDMX REST API for data and structure queries. Delivers SDMX-ML, SDMX-JSON and SDMX-CSV for #4 and #5 datasets.	SAF-AS13	1.5.3 Facilitate data access

ID	Component	Product	Licence	Description	Realises service	Realises capability
SAF-AC14	SDMX Data Explorer	.Stat Suite Data Explorer	MIT (verify per component)	Public web application to search, view, chart and download released SDMX datasets.	SAF-AS14	1.5.3 Facilitate data access
SAF-AC15	BI and Dashboards	Apache Superset	Apache-2.0	Interactive dashboards over released data. Also used for internal quality and KPI dashboards.	SAF-AS15	1.5.4 Statistical information management
SAF-AC16	Open Data Portal	CKAN	AGPL-3.0	Open-data catalogue and bulk download with machine-readable (DCAT) metadata. Links to SDMX dataflows.	SAF-AS16	1.5.1 Publication management
SAF-AC17	Research Workbench	JupyterHub	BSD-3-Clause	Secure containerised notebook environment for accredited researchers and for VS-08 sandboxes. Outputs are checked before they leave the environment.	SAF-AS17	2.2.3 Methodology; 2.2.4 Data services
SAF-AC21	Service Desk	Zammad	AGPL-3.0	User-support ticketing, knowledge base and feedback capture.	SAF-AS21	3.4.2 Contact management; 3.4.4 Self-service

ID	Component	Product	Licence	Description	Realises service	Realises capability
SAF-AC22	Web Analytics	Matomo	GPL-3.0	Privacy-respecting usage statistics for portals, APIs and releases.	SAF-AS22	2.2.5 User research

Security, Access and Delivery

ID	Component	Product	Licence	Description	Realises service	Realises capability
SAF-AC18	Identity and Access Management	Keycloak	Apache-2.0	Single sign-on (OIDC/SAML), multi-factor authentication, and role and group management for staff, providers and researchers.	SAF-AS18	3.3.4 Logical identity and access control
SAF-AC19	Policy Decision Engine	Open Policy Agent	Apache-2.0	Policy-as-code decisions for Trino, APIs and research access, based on role, BIV class and steady state.	SAF-AS19	3.3.4 Logical identity and access control
SAF-AC20	API Gateway	Apache APISIX	Apache-2.0	Single entry point for external APIs: authentication, rate limiting and usage logging.	SAF-AS20	3.3.6 IT security services

ID	Component	Product	Licence	Description	Realises service	Realises capability
SAF-AC23	Source Control and CI/CD	GitLab CE	MIT	Git repositories, merge requests and CI pipelines. System of record for SQL models, DAGs, schemas and policies. Promotes versions from test to production.	SAF-AS23	3.2.1 Application development

Application services

Each application service is realised by one component. Most services realise one or more business services of the value streams.

ID	Application service	Description	Realised by	Realises business services
SAF-AS01	Survey Collection Service	Collect responses through web questionnaires.	SAF-AC01	Data Collection Service
SAF-AS02	Secure File Receipt Service	Receive files from data providers securely.	SAF-AC02	Data Collection Service
SAF-AS03	Data Ingestion Service	Acquire data from APIs, databases and web sources into the raw zone.	SAF-AC03	Data Collection Service
SAF-AS04	Pipeline Orchestration Service	Schedule, run, monitor and retry pipelines; expose run status.	SAF-AC04	Ingestion Monitoring Service
SAF-AS05	SQL Query and Processing Service	Run SQL over Iceberg tables for processing and analysis.	SAF-AC05	Raw Data Provisioning Service; Standardization Service; Error-Detection Service; Derivation and Integration Service; Estimation and Aggregation Service; Integration and Composition Service

ID	Application service	Description	Realised by	Realises business services
SAF-AS06	Table Catalog Service	Manage tables, snapshots, tags and quality versions.	SAF-AC06	Raw Data Provisioning Service; Quality-Version Catalog Service; Output-Version Catalog Service
SAF-AS07	Versioned Transformation Service	Execute version-controlled SQL transformations.	SAF-AC07	Standardization Service; Classification Mapping Service; Editing and Correction Service; Derivation and Integration Service; Integration and Composition Service; Indicator and Index Service; Dissemination Preparation Service
SAF-AS08	Statistical Processing Service	Run imputation, weighting, estimation and model workloads.	SAF-AC08	Editing and Correction Service; Imputation Service; Weighting and Calibration Service; Estimation and Aggregation Service; Modeling and Analysis Service; Indicator and Index Service
SAF-AS09	Disclosure Control Service	Assess disclosure risk and apply protection.	SAF-AC09	Validation and Disclosure-Control Service; Dissemination Preparation Service
SAF-AS10	Data Validation Service	Validate data against expectation suites and record results.	SAF-AC10	Ingestion Monitoring Service; Conformance Validation Service; Error-Detection Service; Validation and Disclosure-Control Service; Coherence Validation Service

ID	Application service	Description	Realised by	Realises business services
SAF-AS11	Metadata and Lineage Service	Search, govern and trace datasets and their lineage.	SAF-AC11	Data Agreement Management Service; Metadata Catalog Service; Semantic Harmonization Service; Quality-Version Catalog Service; Output-Version Catalog Service; Product Catalog Service
SAF-AS12	SDMX Structure Registry Service	Maintain and publish SDMX structures, codelists and dataflows.	SAF-AC12	Classification Catalog Service; Classification Mapping Service; Semantic Harmonization Service; Output-Data Provisioning Service; Output-Version Catalog Service; Product Catalog Service; Product Provisioning Service; Dissemination Preparation Service
SAF-AS13	SDMX Data Dissemination Service	Serve SDMX data and structure queries.	SAF-AC13	API and Dashboard Service
SAF-AS14	Data Exploration Service	Explore, chart and download released data.	SAF-AC14	Publication and Open-Data Service
SAF-AS15	Dashboard Service	Provide dashboards and visual analytics.	SAF-AC15	API and Dashboard Service; Analytical Content Service
SAF-AS16	Open Data Catalog Service	Publish open data with machine-readable metadata.	SAF-AC16	Publication and Open-Data Service
SAF-AS17	Research Workbench Service	Provide secure analysis environments.	SAF-AC17	Analytical Content Service; Secure Research Access Service
SAF-AS18	Authentication Service	Authenticate users and services.	SAF-AC18	Secure Research Access Service
SAF-AS19	Authorization Service	Decide access by policy.	SAF-AC19	Secure Research Access Service
SAF-AS20	API Management Service	Expose, secure and meter APIs.	SAF-AC20	API and Dashboard Service

ID	Application service	Description	Realised by	Realises business services
SAF-AS21	User Support Ticketing Service	Handle user questions and feedback.	SAF-AC21	User Support Service; Usage and Feedback Service
SAF-AS22	Usage Analytics Service	Measure usage of releases and services.	SAF-AC22	Usage and Feedback Service
SAF-AS23	Version Control and CI/CD Service	Version, review, test and promote design artefacts.	SAF-AC23	Design and Methodology Service
SAF-AS24	Dataset Registration Event Service	Signal that a new steady-state version is registered, which triggers the next stream (asset update plus Kafka message).	SAF-AC04	Raw Data Provisioning Service; Standardized Data Provisioning Service; Processed-Data Provisioning Service; Output-Data Provisioning Service; Product Provisioning Service

Application interfaces

Five application services are exposed through an explicit interface. Each interface is part of the component that realises the service.

ID	Application interface	Description	Part of	Exposes service
SAF-AI01	SQL Interface (JDBC/ODBC/HTTP)	Trino client protocol for analysts, BI tools and pipelines.	SAF-AC05	SAF-AS05
SAF-AI02	SDMX REST API	SDMX data and structure queries (SDMX-ML, SDMX-JSON, SDMX-CSV).	SAF-AC13	SAF-AS13
SAF-AI03	Iceberg REST Catalog API	Standard Iceberg REST catalog endpoint used by Trino, dbt and Spark-compatible clients.	SAF-AC06	SAF-AS06
SAF-AI04	Metadata and OpenLineage API	Ingest and query metadata and lineage events.	SAF-AC11	SAF-AS11
SAF-AI05	Orchestration REST API	Trigger and monitor pipeline runs; asset events.	SAF-AC04	SAF-AS04

Application services per value stream

The application services serve the business processes of the six business domains. The table shows which services serve which process type of each primary value stream.

PLACEHOLDER: ArchiMate view *Phase C - Application to Value Stream Mapping*

Value stream	Design and Implementation	Execution	Management
VS-01 Data Acquisition and Ingestion	SAF-AS23	SAF-AS01, SAF-AS02, SAF-AS03, SAF-AS04, SAF-AS06, SAF-AS10, SAF-AS11, SAF-AS24	SAF-AS11
VS-02 Data Standardization and Integration	SAF-AS23	SAF-AS04, SAF-AS05, SAF-AS06, SAF-AS07, SAF-AS10, SAF-AS11, SAF-AS12, SAF-AS24	SAF-AS11
VS-03 Data Editing and Enrichment	SAF-AS23	SAF-AS04, SAF-AS05, SAF-AS06, SAF-AS07, SAF-AS08, SAF-AS10, SAF-AS11, SAF-AS24	SAF-AS11
VS-04 Estimation, Analysis, and Statistical Modeling	SAF-AS23	SAF-AS04, SAF-AS05, SAF-AS06, SAF-AS08, SAF-AS09, SAF-AS10, SAF-AS12, SAF-AS24	SAF-AS11
VS-05 Statistical Product Composition	SAF-AS23	SAF-AS04, SAF-AS05, SAF-AS06, SAF-AS07, SAF-AS08, SAF-AS10, SAF-AS12, SAF-AS15, SAF-AS17, SAF-AS24	SAF-AS11
VS-06 Dissemination and Service Delivery	SAF-AS23	SAF-AS09, SAF-AS12, SAF-AS13, SAF-AS14, SAF-AS15, SAF-AS16, SAF-AS17, SAF-AS18, SAF-AS19, SAF-AS20, SAF-AS21, SAF-AS22	SAF-AS11, SAF-AS22

Four services return in almost every stream: orchestration (SAF-AS04), the table catalog (SAF-AS06), validation (SAF-AS10) and the registration event (SAF-AS24). They are the shared mechanics of a steady-state handover: run the pipeline, validate against the quality gate, tag the version, and signal the next stream.

Core data platform and SDMX path

The core of the landscape is the path that data follows from #1 to #5. Each steady-state dataset is accessed by a small set of components and stored in a zone of its own.

PLACEHOLDER: ArchiMate view *Phase C-D - Core Data Platform and SDMX Path*

Steady-state dataset	Components that access it	Stored in
Raw Dataset (#1)	Survey Platform, Secure File Transfer Gateway, Ingestion Engine, Data Quality Framework, Distributed SQL Query Engine	Raw Zone
Standardized Dataset (#2)	SQL Transformation Framework, Distributed SQL Query Engine, Data Quality Framework	Standardized Zone
Processed Dataset (#3)	SQL Transformation Framework, Statistical Computing Runtime, Distributed SQL Query Engine, Data Quality Framework	Processed Zone
Statistics Dataset - SDMX (#4)	Statistical Computing Runtime, Statistical Disclosure Control Toolkit, Distributed SQL Query Engine	Statistics Zone, SDMX Messages
Statistical Product - SDMX Dataflow (#4)	SQL Transformation Framework, SDMX Structure and Lifecycle Manager	
Released Dataset - SDMX (#5)	Statistical Disclosure Control Toolkit, Distributed SQL Query Engine, SDMX Web Service, SDMX Data Explorer, BI and Dashboards, Open Data Portal, Research Workbench	Released Zone, SDMX Messages